

CODE SPORT



In this month's column, we feature a set of computer science interview questions.

We have been discussing natural language processing techniques in the last few columns. However, given that this is the first edition of CodeSport for 2016, let's follow our tradition of welcoming the New Year with a bunch of computer science interview questions — focusing on algorithms, data structures, machine learning and natural language processing.

- (1) For dengue fever, a diagnostic test has been developed. The new test is 81 per cent reliable (i.e., it misses 19 per cent of cases where the patient actually has the disease). It can wrongly label a patient as having dengue fever in 10 per cent of cases when the patient actually does not have the disease. You are told nothing about the patient's symptoms or any other clinical information, except that the patient has tested positive for the dengue diagnostic test. Can you figure out whether the patient actually has dengue from the above given information? If yes, how? If not, why not?
- (2) Consider the above question. Now you are given the additional information that in the patient's city, dengue fever affects only 1 per cent of the total population. Can you now compute the probability of whether the patient actually has dengue fever or not?
- (3) You are given two arrays and asked to find the common elements between the two arrays. (a) Assume that the two arrays only contain integer elements. (b) Assume that the two arrays contain floating point values. Would your solution for (a) work for (b) as well?
- (4) You are given a coin and after tossing it 10 times, you observe that you get heads in eight out of the 10 tosses. (a) How can you determine whether the given coin is fair or not? (b) Now assume that you are given 10 coins and asked to identify the coins which are not fair. How would you modify your solution for (a) to solve the problem given in (b)?
- (5) You are given an array A of N integers. You are also given a number X. You are asked to find the element in A that is closest to X. You are given the constraint that you should not sort the array A. (a) Can you write a C program to solve this problem? (b) Now you are told that the set S consists of floating point values instead of integers. How would you modify your solution for (a) to solve (b)?
- (6) In solving classification and regression problems in machine learning, great emphasis is laid in feature engineering, where the right set of features are carefully selected among all the features of the given data samples. For instance, consider a linear regression problem where each sample of input data has four features: X_1 , X_2 , X_3 and X_4 . X_3 and X_4 are linearly dependent on X_2 . X_1 is linearly independent of all the other features. For example, assume that the regression problem is to predict house prices. X_1 is the type of locality, X_2 is the size of the house, X_3 and X_4 are the sizes of the bedroom and size of the external yard, respectively. You have created two classifiers, G1 and G2 such that G1 uses all four features and G2 uses only X_1 and X_2 . Will G2 be more accurate than G1? What are the problems with G1 in comparison with G2?
- (7) What is cross-validation? Given a data set of 10,000 samples, how would you divide it into training, validation and test data sets? How would you perform cross validation on the given data set?
- (8) Tagging parts of speech (PoS) is an important problem in natural language processing, where the tokens in a given sentence are to be annotated with their corresponding parts of speech tags. For example, consider the following sentence: John ate the apple. This will be tagged as follows: John<NP> ate <VP> the <DT> apple <NP>, where NP stands